WHITE PAPER / PROPOSAL FOR EXPERIMENTAL EVALUATION

One Collective Reference, Two Extensive Coordinates

A percolation response to Forbes's proposal for atomic-level constants and units

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Abstract

This paper proposes a shared experimental reference for nanoparticle amount and electric charge, anchored to the onset of a conducting network. It responds to Richard G. Forbes's call for fuller atomic-level nomenclature and to the earlier entity unit proposed by B. P. Leonard. A charge-selected particle stream would deposit a counted amount while delivering a measured charge; percolation would identify a reproducible collective endpoint. For a fixed reference count, paired amount and charge scales follow from the existing constants N_A and e. The proposed advantage is operational parsimony: one preparation, reference cell and endpoint serving two extensive quantities. This is a hypothesis about laboratory practice, not a reduction in the number of SI dimensions or a new charge quantum. The paper specifies the required particle-to-charge bridge, an uncertainty model and tests that could establish or reject the proposed advantage.

1 The proposal in one statement

Investigate one percolation experiment as a common physical reference for two additive quantities: the amount of specified nanoparticles deposited and the electric charge delivered with them.
Retain the exact SI definitions. Treat the collective endpoint as a measured material characteristic and, if useful, assign a fixed reference count near that endpoint for paired working scales.

The motivating question is whether atomic-level reference practice can do more than attach names to individual counting units. An assembled material can acquire a measurable conducting pathway as its particle loading increases. If the arriving particles also carry a characterised charge, the same preparation could connect particle amount, delivered charge and the appearance of collective function.

The strongest defensible claim is therefore *one reference procedure with two extensive outputs*. A universal transition count is neither assumed nor required. The proposed combination and its claimed efficiency have not been demonstrated in this paper.

PROVENANCE AND MOTIVATION

2 An Open table

Leonard's 2007 article [1] and Forbes's 2020 article on field-desorption theory [2]. Leonard's paper is the earlier source for the name *entity* and symbol ent. His wider pre-2019 proposals should be read in their historical context.

Poster entry	**Mass**	**Electric charge**	**Amount of substance**
Atomic-level constant	Unified atomic mass constant	Elementary positive charge	Name open
Constant symbol	m_{u}	e	n_1 proposed
Atomic-level unit name	Dalton / unified atomic mass unit	No specific name	Entity proposed
Unit symbol	Da / u	No specific symbol	Symbol open

Table 1: Historical status of standards audit.

The charge gaps concern the *name and symbol of an atomic-level unit*. The elementary charge is already present. A dimensionless charge number $Z = Q/e$ is also mathematically available; the table does not establish that a charge number is physically missing.

Forbes's elementary amount is the reciprocal of Avagadro constant

$$n_1 = \frac{1}{N_{\mathrm{A}}}, \qquad n = N n_1. \tag{1}$$

The present proposal accepts this relation. It asks whether one collective reference procedure can support both the amount and charge columns.

The SI advantage that should be retained

The current SI fixes the numerical values of N_{A} and e exactly.

$$N_{\mathrm{A}} = 6.02214076 \times 10^{23}\,\mathrm{mol}^{-1},$$
$$e = 1.602176634 \times 10^{-19}\,\mathrm{C}.$$

Its definitions are separated from particular material specimens and laboratory implementations [3, 4]. Thus, giving n_1 a name is compatible with the existing definition of amount. A percolation endpoint has additional material and geometry dependencies.

The distinction is essential: *a unit definition, a reference specimen and a measured transition are different metrological objects*. The reference proposed here would sit beneath the SI as a possible working practice. It would not replace the mole or coulomb.

QUANTITIES BEFORE UNITS

3 An extensive number and charge framework

Let N_p count a specified kind of nanoparticle. It counts particles, including a stated rule for aggregates, rather than the atoms inside each particle. The associated amount is

$$n_\mathrm{p} = \frac{N_\mathrm{p}}{N_\mathrm{A}}, \qquad c_\mathrm{p} = \frac{N_\mathrm{p}}{V}. \tag{2}$$

Particle number and amount add across disjoint inventories. Number density is intensive under replication of the same material state.

For a selected particle phase with particle charge numbers z_i,

$$Q_\mathrm{p} = e \sum_{i=1}^{N_\mathrm{p}} z_i. \tag{3}$$

For a uniform incoming charge state z, the charge delivered by ideal complete collection is

$$Q_\mathrm{dep} = zeN_\mathrm{p} = zeN_\mathrm{A}n_\mathrm{p}. \tag{4}$$

This is the required bridge. Its application depends on verified charge state, collection and charge accounting.

Quantity	SI	Role
Particle count N_p	Unit one	Additive inventory of specified particles
Particle amount n_p	mol	Extensive amount of those particles
Particle-phase charge Q_p	C	Signed charge inventory
Deposited charge Q_dep	C	Cumulative charge delivered during collection
Network conductance G	S	Geometry-dependent collective response

Table 2: A charge inventory, a delivered charge and a conductance measurement are kept distinct.

Extensivity and its limits

For independent deposition histories, particle counts and delivered charges add. Combining M identical reference runs gives M times each quantity. The resulting conductance need not be M times its original value: electrical connection and specimen geometry must also be specified.

The count of particles belonging to a spanning network can be informative, but connecting two clusters can change membership without adding particles. It should be reported as a structural observable, with bulk-scaling and boundary qualifications, rather than used as an unconditional counting standard [5].

An electrically neutral assembly can percolate. Neither charge neutrality nor conduction onset establishes equation (4); incoming charge selection provides the additional information.

A COLLECTIVE ENDPOINT WITH A SPECIFIED PROTOCOL

4 Define the percolation reference cell

Conducting paths arise as particles become sufficiently connected through contacts, tunnelling or hopping. Thresholds depend on particle geometry, interactions and matrix conditions [6]. Conductance has been monitored during deposition of percolative metal nanoparticle films, providing a relevant experimental precedent [7].

Specify a protocol $\mathcal{P}$: material, particle size distribution, surface chemistry, substrate, collector geometry, electrode spacing, temperature, atmosphere, deposition history and readout bias. In a finite specimen define an operational median onset by

$$\Pr(G \geq G_{\min} \mid N_{\mathrm{p}} = N_{50}, \mathcal{P}) = \frac{1}{2}. \tag{5}$$

Here $G_{\min}$ and the persistence time are agreed measurement criteria. This is a finite-specimen reproducibility convention. It is not automatically the infinite-system geometrical percolation threshold.

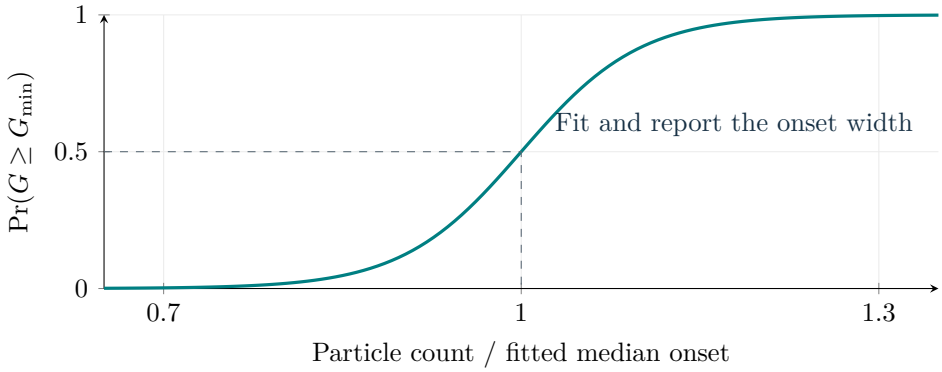


Figure 1: Illustrative logistic curve only; no measured data. The spread of onset counts is an outcome to be assessed, rather than suppressed by calling the endpoint exact.

For sufficiently large comparable bulk specimens one may expect $N_{50} \approx c_{\star}V$, with finite-size corrections. For thin deposited films, an areal loading and a fixed electrode pattern are often more appropriate. Repeating the same cell provides the cleanest additivity test; changing its shape is a separate experiment.

Percolation supplies a functional endpoint: the onset of a conducting network. An absorptivity criterion such as 0.99 could provide an optical working reference, but would normally select a point on a smooth response curve and would require wavelength and geometry specifications.

ONE DEPOSITION HISTORY, TWO MEASURANDS

5 Make the particle-to-charge bridge physical

The most direct proof-of-principle design would use size-selected, charge-characterised incoming nanoparticles. For a verified single positive charge state, each collected particle contributes $+e$ to the delivered charge budget. Established aerosol metrology already connects singly charged particle flux to electrical current [8, 9]. The new proposal is to couple that practice to a percolation endpoint.

Proposed implementation. Deposit charge-selected conducting nanoparticles into a guarded collector containing a reference cell. Interleave deposition and low-bias conductance readings. Integrate the corrected collection current up to the agreed endpoint, and independently check the deposited particle inventory.

Let $I_{\mathrm{col,corr}}$ include the necessary current, leakage, secondary-emission and charge-storage corrections. Then, under the verified ideal collection model,

$$Q_{\mathrm{dep}}(t) = \int_0^t I_{\mathrm{col,corr}}(t')\,\mathrm{d}t', \qquad N_{\mathrm{p}}(t) = \frac{Q_{\mathrm{dep}}(t)}{ze}. \tag{6}$$

For $z = 1$, the endpoint outputs are the paired estimates $\widehat{n}_{50} = \widehat{N}_{50}/N_{\mathrm{A}}$ and $\widehat{Q}_{\mathrm{dep},50} = e\widehat{N}_{50}$. Their common information should be explicit.

Three conditions the experiment must establish

1. **Identity survives collection.** A counted incoming cluster corresponds to the specified deposited entity, or fragmentation and coalescence are quantified.
2. **Collected charge is accounted for.** Neutrals, multiply charged particles, lost particles, emitted electrons and residual substrate charging are excluded or corrected with stated uncertainty.
3. **Readout current is separate.** Charge flowing through the developing film under bias is distinguished from arriving particle charge. Interleaving alone must be validated against capacitive transients.

In particular, $Q_{\mathrm{tr}} = \int I_{\mathrm{film}}\,\mathrm{d}t$ can greatly exceed eN_{p}: the same network can carry successive electrons. A conducting endpoint therefore does not imply one transported electron per nanoparticle.

The collector and charging design remain experimental development tasks. The cited studies establish component precedents; they do not demonstrate this complete joint reference.

FIXED DEFINITIONS AND A CONDITIONAL PARSIMONY CLAIM

6 Two working scales from one reference count

First estimate N_{50} and its variability. If comparisons justify a working reference, adopt a fixed integer N_{ref} near the agreed endpoint. It may differ slightly from a later, improved estimate of N_{50}. Define paired reference amounts

$$\boxed{n_{\mathrm{P}} = \frac{N_{\mathrm{ref}}}{N_{\mathrm{A}}}, \qquad q_{\mathrm{P}} = N_{\mathrm{ref}} e.} \tag{7}$$

The charge scale uses the single-charge reference case. A known charge state z delivers zq_{P} for N_{ref} particles. If working unit names are later desired, *percolation entity* could denote n_{P}; a distinct charge designation would be needed for q_{P}. At this stage both should be reported in mol and C.

The corresponding numerical coordinates are

$$X_n = \frac{n_{\mathrm{p}}}{n_{\mathrm{P}}} = \frac{N_{\mathrm{p}}}{N_{\mathrm{ref}}}, \qquad X_Q = \frac{Q}{q_{\mathrm{P}}} = \frac{Q/e}{N_{\mathrm{ref}}}. \tag{8}$$

Both are additive for their specified inventories or process histories. They are not sample-dependent fractions of an independently recalculated threshold. Redefining N_{ref} for every specimen would discard the common scale.

The elementary ent discussed by Leonard and Forbes remains: $n_{\mathrm{P}} = N_{\mathrm{ref}}\, n_1$. Thus the proposal extends a reference practice around his elementary quantity rather than changing its conversion to the mole. The exact assigned scales satisfy

$$\frac{q_{\mathrm{P}}}{n_{\mathrm{P}}} = eN_{\mathrm{A}} = F. \tag{9}$$

The Faraday relation is already present in the poster. Neither this ratio nor the single-charge counting principle is claimed as a new discovery.

What two uses for one reference can mean

One count multiplier, one cell specification and one deposition endpoint organise amount and charge together. A physical collection model gives the connection experimental content. This is a candidate saving in preparation, documentation and calibration workflow.

It does not reduce two physical dimensions to one, eliminate independent validation, or supply a new fundamental constant. Once N_{ref} is fixed, equation (7) is exact by convention; correspondence with percolation retains experimental uncertainty.

Illustrative arithmetic, not data. If $N_{\mathrm{ref}} = 10^9$, then $n_{\mathrm{P}} \approx 1.66054\,\mathrm{fmol}$ and $q_{\mathrm{P}} = 160.2176634\,\mathrm{pC}$. These values illustrate the common multiplier; 10^9 has not been established as a transition count.

HOW THE PROPOSAL WOULD EARN ITS CLAIM

7 Validation, uncertainty and a decision rule

A minimum comparison programme

1. **Establish a baseline.** Document separate particle-amount and charge procedures at agreed target uncertainties.
2. **Qualify the stream and collector.** Measure the charge-state distribution, collection losses and net charge recovery. Characterise particle size and aggregation before and after deposition.
3. **Locate the endpoint.** Use replicate reference cells across a loading range; fit equation (5) and report its width, batch dependence and preparation sensitivity.
4. **Break the circularity.** Validate the charge-derived particle inventory against an independent counting or concentration method. Agreement between two outputs computed from the same current integral is not independent evidence.
5. **Test replication.** Compare summed inventories and charges from independent cells. Then vary geometry and state separately to establish the limits of transfer.
6. **Audit parsimony.** Compare preparation time, reference artefacts, calibration steps, correction parameters and maintained uncertainties against the baseline.

For separately obtained estimates of delivered charge and deposited count, a diagnostic bridge factor is

$$\kappa = \frac{Q_{\mathrm{dep}}}{eN_{\mathrm{p}}}, \qquad u_r^2(\kappa) \simeq u_r^2(Q_{\mathrm{dep}}) + u_r^2(N_{\mathrm{p}}) - \frac{2\ \mathrm{cov}(Q_{\mathrm{dep}}, N_{\mathrm{p}})}{Q_{\mathrm{dep}}\, N_{\mathrm{p}}}. \tag{10}$$

This first-order expression assumes positive, nonzero quantities in the selected positive-charge experiment. The elementary charge is exact in SI; uncertainty enters the realised quantities and correction model. For the ideal single-charge collection model, $\kappa = 1$. Apply a full uncertainty budget where linear propagation is inadequate [10].

Decision rule

Claim a practical parsimony gain only if the joint procedure achieves the pre-agreed amount and charge uncertainties with lower documented total burden. Charge selection, microscopy validation and collector corrections must count as part of that burden.

A stable κ with reproducible onset would support a joint working reference. A variable charge-state bridge, unstable threshold, or a greater total calibration burden would reject the claimed advantage for that material system. Either outcome can improve nanoparticle characterisation without changing the SI.

POSITION AND REFERENCES

8 Recommended standards position

Forbes's proposal has the stronger immediate claim to exact, universal atomic-level nomenclature: its elementary amount follows directly from the defined N_A. The present proposal offers an additional research question about shared experimental reference practice.

The recommended route is a published protocol and comparison study. Retain the mole, coulomb, e and N_A; distinguish assigned reference scales from measured endpoints; and standardise terminology only if the joint method proves useful. The proposed parsimony is one collective endpoint serving two extensive coordinates, with all necessary charge accounting visible.

Evidence status. The historical citations, SI relationships, charged-aerosol precedent and percolative-film precedent are sourced below. The combined collector design, reference assignment, comparison programme and efficiency claim are proposals made here.

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